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A DIGITAL WALKIE TALKIE ARCHITECTURE UTILIZING GFSK

EC 256 - Communication Lab Project

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Project Overview

- System Introduction -Work Flow of The Device
- Electret microphone
- Audio Pre-Amplification - Circuit workflow & verification
- GFSK Modulation- Circuit workflow & verification
- ADC(Analog to Digital Converter) - Theoretical vs Practical Analysis, result
- Transmitter & Receiver
- DAC(Digital to Analog Converter) - Theoretical vs Practical Analysis, result
- GFSK Demodulation
- Audio Power Amplification
- Soucres

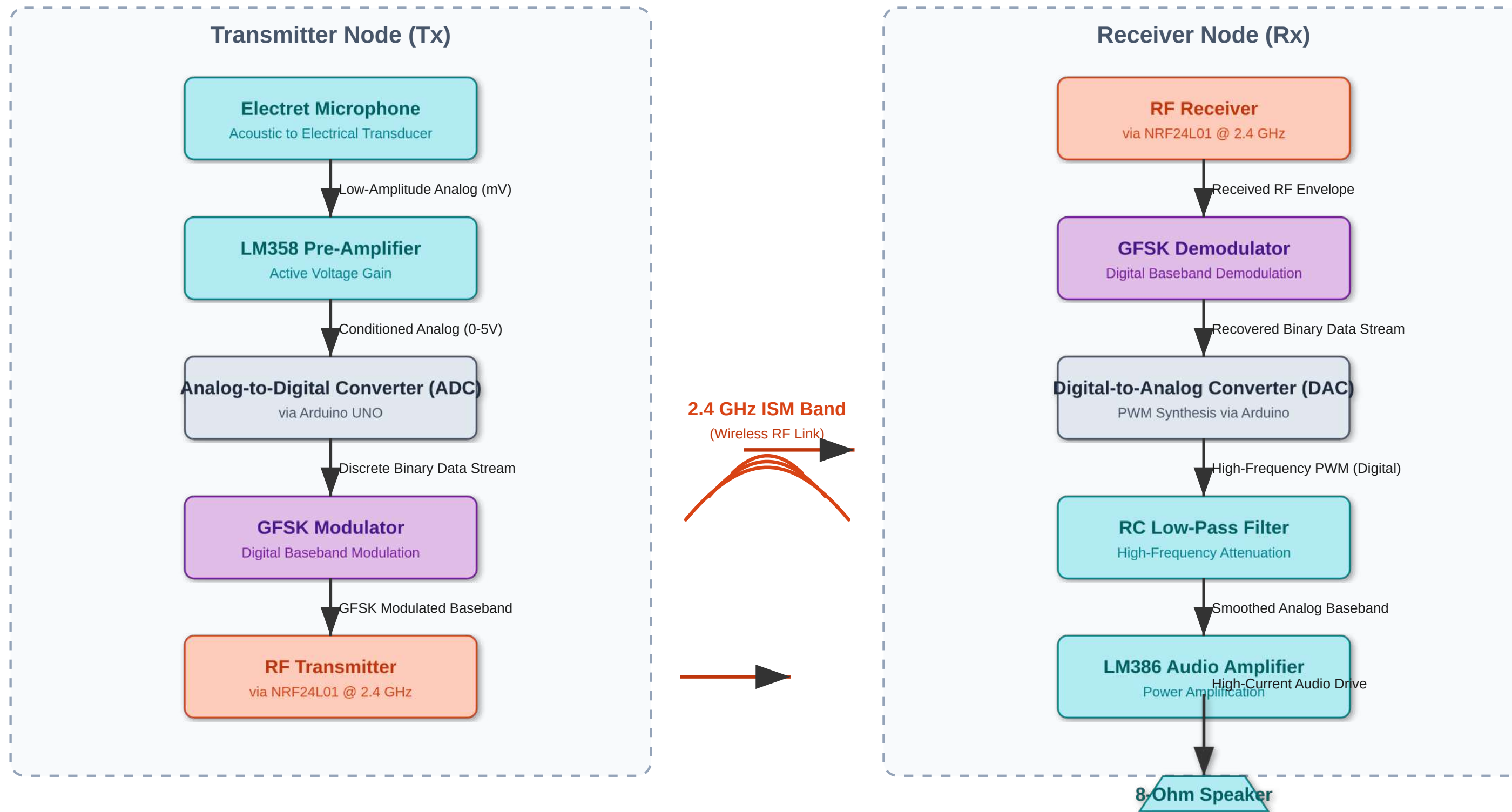
SYSTEM INTRODUCTION

Workflow of the Device

Block Diagram - Work Flow of The Device

System Architecture: Digital Voice Transceiver via GFSK Modulation

(Half-Duplex Operation)



ELECTRET MICROPHONE

Electret Microphone

Technical Aspect:

- An electret microphone is an electrostatic transducer that uses a permanently polarized dielectric material, eliminating the need for a high-voltage external power supply.
- It features a built-in Field Effect Transistor (FET) to convert high-impedance signals into a stable output suitable for our pre-amplifier.

Working Principle:

- **Vibration:** Incoming sound waves cause a thin, flexible diaphragm to vibrate against a fixed backplate.
- **Capacitance Change:** This movement varies the distance between the plates, creating a proportional change in electrical capacitance.
- **Signal Conversion:** The permanently stored charge on the electret material converts these capacitance changes into a fluctuating AC voltage.

Why This Microphone?

- **High Sensitivity:** It captures crisp, detailed audio even at low speaking volumes, which is essential for clear walkie-talkie communication.
- **Compact Durability:** The small, rugged form factor is ideal for fitting into a handheld, portable device enclosure.
- **Power Efficiency:** It operates on minimal current, significantly extending the life of our 9V battery during long-term use.

AUDIO PRE- AMPLIFICATION

Circuit Workflow and Verification

Audio Pre-Amplification

What it is: A specialized electronic circuit designed to prepare a weak electrical signal for further processing or transmission.

What it does: It boosts the tiny millivolt output of the microphone to a stable "line-level" signal while keeping background hiss to a minimum.

How it works:

- **Transduction:** The electret microphone converts physical sound waves into a very low-voltage AC electrical signal.
- **DC Isolation:** Coupling capacitors allow the AC voice signal to pass through while blocking DC battery voltage from disrupting the transistor.
- **Active Amplification:** The LM358 uses high-gain differential amplification to increase the signal amplitude based on the ratio of external resistors. By using capacitors in the feedback loop, the pre-amp can selectively amplify voice frequencies while rolling off low-frequency hum and high-frequency hiss.
- **Signal Replication:** This process creates an exact, high-voltage replica of the original sound wave at the output terminal.

Circuit Workflow

- 1. Mic Biasing (R1)** – R1 supplies the DC bias voltage required to power the electret microphone capsule.
- 2. Input Coupling (C1)** – Blocks DC from the mic and passes only the AC audio signal forward into the amplifier stage.
- 3. Input Impedance Setting (R2)** – Sets the input impedance seen by the signal before entering the inverting input (pin 2) of the LM358.
- 4. Virtual Ground Biasing (R3, R4)** – R3 and R4 form a voltage divider that sets the non-inverting input (pin 3) to $V_{cc}/2$, establishing a stable mid-rail DC operating point for single-supply operation.
- 5. Signal Amplification (IC1 – LM358)** – The op-amp amplifies the audio signal in inverting configuration, with gain precisely determined by the ratio of R5 to R2.
- 6. Gain Setting (R5)** – Acts as the feedback resistor; the voltage gain is set by $-R5/R2$, giving predictable and stable amplification.
- 7. Supply Decoupling (Pin 8, Pin 4)** – Pin 8 connects to Vcc and pin 4 to GND, powering the op-amp with a clean single-supply rail.
- 8. Output Coupling (C2)** – Blocks the DC offset from the op-amp output and delivers a clean AC audio signal to the next stage

GFSK MODULATION

Circuit Workflow and Verification

GFSK Modulation

A GFSK modulator is a type of frequency modulation where the digital data is first passed through a Gaussian filter before modulating the carrier frequency. This smoothing reduces sudden frequency transitions, making the signal more bandwidth-efficient and less noisy.

Advantages of GFSK

1. Bandwidth Efficient - Gaussian filtering reduces sidebands
2. Reduced Interference - Lower adjacent channel interference
3. Smooth Phase Transitions - No abrupt phase changes → better signal quality
4. Power Efficient - Constant envelope signal → efficient for RF amplifiers

ANALOG TO DIGITAL CONVERTER

Arduino Internal ADC

The Arduino's internal ADC is a built-in 10-bit converter that turns your voice into digital data without needing extra chips or messy wiring. It uses an internal "Sample-and-Hold" circuit to capture audio snapshots accurately while saving space and battery power.

Advantages of ADC

1. Compact Design
2. Power Efficiency
3. Reduced Noise
4. Lower Latency

TRANSMITTER AND RECEIVER

The Packets

What it does: After the Logic being converted to 3.3V CMOS logic.

The problem: The Transmitter and Receiver have input and output Bottlenecks at 3.3v. And the input to the Transmitter has to be a digital signal.

The Microcontroller: The microcontroller is responsible to 'Packet' the data allowing up to 32 bytes per packet.

The Channel: The sensor (NRF24L01) uses SPI (Serial Peripheral Interface) with 4 pins (MISO, MOSI, SCK, CSN) to connect to a microcontroller.
used as both the receiver and a transmitter . the module can send data in "packets"

Circuit & Component Decisions

Stage 1 - Arduino holds the incoming data as buffer in it's memory.

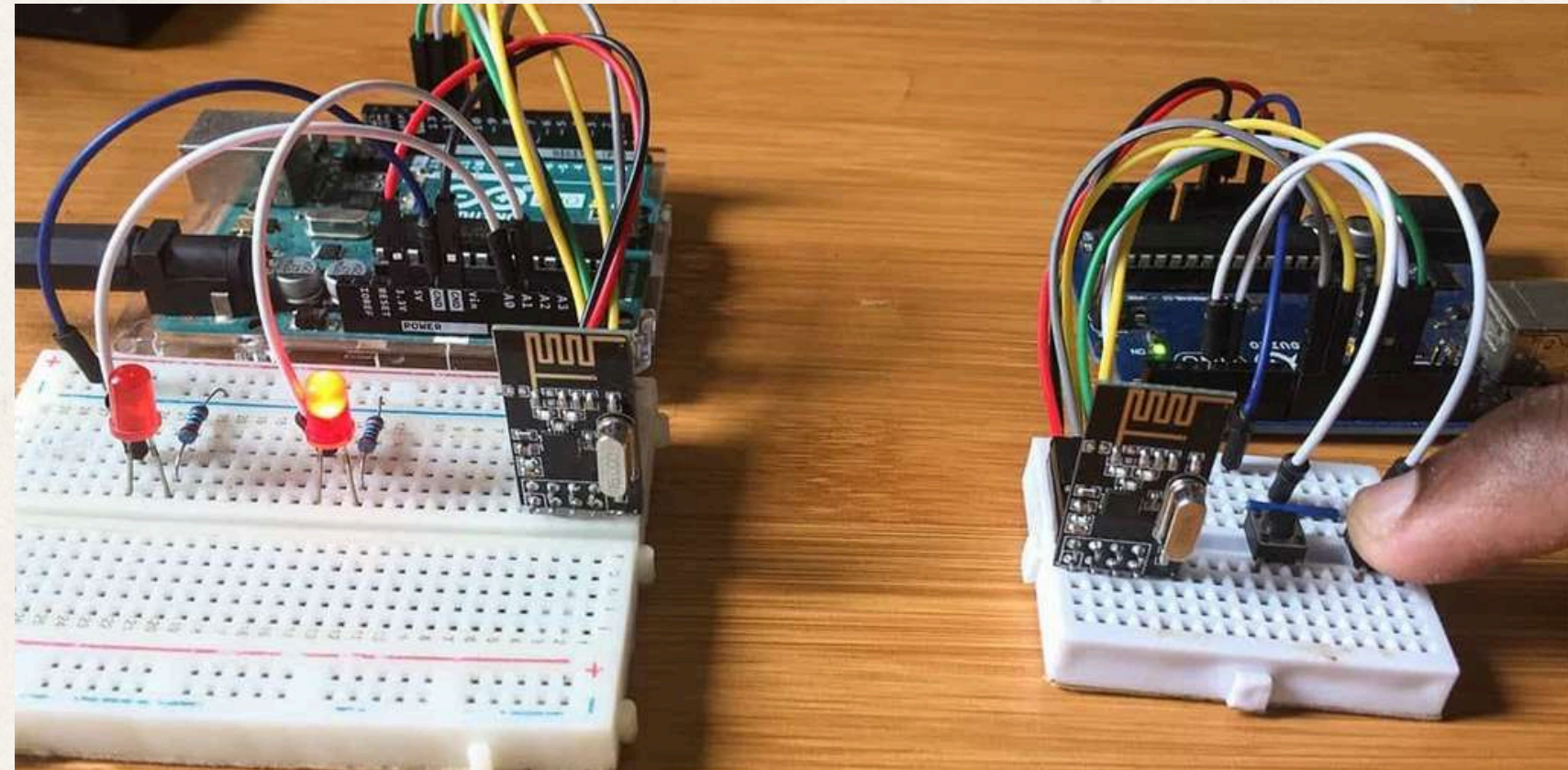
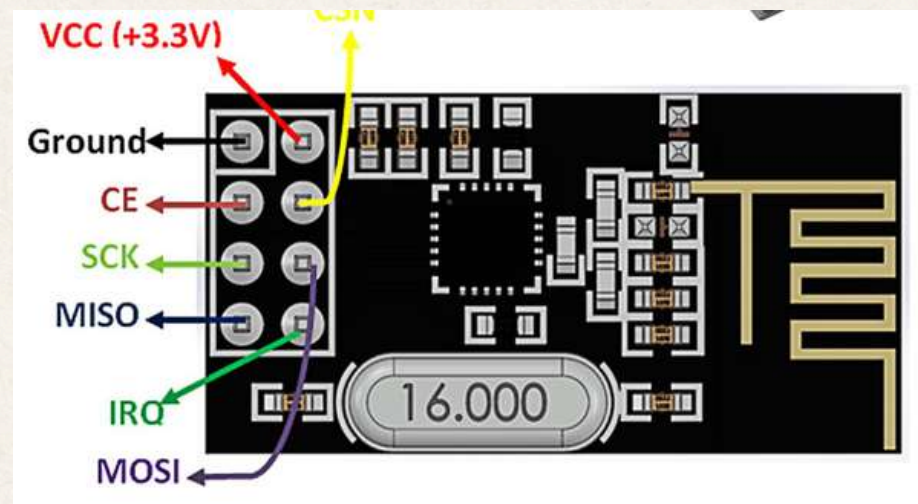
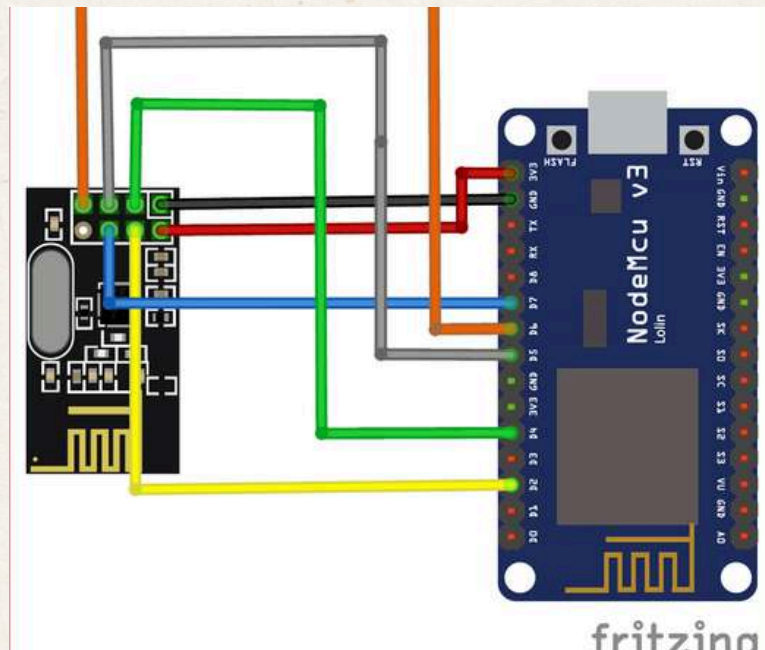
Stage 2 - After 32 Bytes of audio data collected, It packets this data.

Stage 3 - Arduino sends this data to the Transmitter

Stage 4 - The Data packets reach the receiver and It stores the data as buffer in the other Arduino.

Stage 5 - Arduino streams this digital signal into the ADC

Circuit Verification - Theoretical vs Practical Analysis



GFSK DEMODULATION

GFSK Demodulator and its Working

GFSK demodulation is the process of recovering digital data from a GFSK-modulated signal. In GFSK, the input bits are first passed through a Gaussian filter (to smooth transitions), then used to frequency-modulate a carrier. The demodulator reverses this process by detecting frequency variations and reconstructing the original data.

Working:

- Received GFSK signal enters Bandpass Filter
- Limiter removes amplitude variations
- Signal goes to Frequency Discriminator
- Frequency deviations converted into voltage
- Low Pass Filter smooths the signal
- Comparator recovers digital data

Advantages of GFSK Demodulator

1. Better Noise Immunity

- Since information is in frequency, it is less affected by amplitude noise

2. Reduced Interference

- Gaussian filtering reduces bandwidth → less adjacent channel interference

3. Efficient for Wireless Systems

4. Power Efficient

- Works well with non-linear power amplifiers

5. Lower Bit Error Rate (BER)

- Smooth transitions reduce sudden changes → better detection

DIGITAL TO ANALOG CONVERTER

Arduino Internal DAC

The new DAC is an internal peripheral built directly into the Arduino. Unlike the external MCP4921, which "talks" to the Arduino over wires, this one is part of the processor itself, allowing it to convert digital audio data into sound waves instantly.

Advantages of DAC

1. Zero-Wire Connection
2. Faster Audio Response
3. Battery Saver
4. Cleaner Signal

LOW PASS FILTER

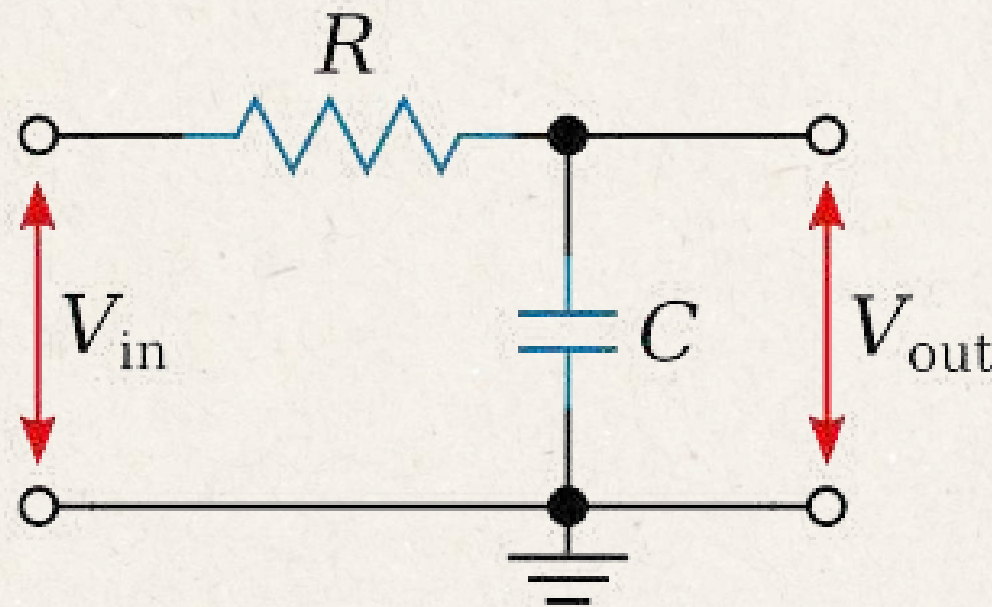
Low Pass Filter

A Low Pass Filter (LPF) allows low-frequency signals to pass and blocks high-frequency components. In communication circuits (like GFSK demodulation), it helps recover the actual data signal from a noisy or rapidly varying waveform.

LPF removes this noise and keeps only useful data.

Circuit Elements:

- Resistor (R)
- Capacitor (C)



AUDIO POWER AMPLIFICATION

Audio Power Amplification

- **What it is:** The final stage of the audio chain designed to boost the signal's current and power rather than just its voltage.
- **What it does:** It provides the necessary "muscle" to drive a low-impedance load, such as a speaker, converting electrical energy into audible sound waves.

How it Works

- **Input Conditioning:** The signal passes through a potentiometer to control the volume level entering the IC.
- **Internal Gain Scaling:** The LM386 uses a specific capacitor between Pin 1 and Pin 8 to set the internal gain to 200x.
- **Impedance Matching:** The circuit translates the high-impedance signal into a low-impedance output suitable for the speaker.

Role of Key Components

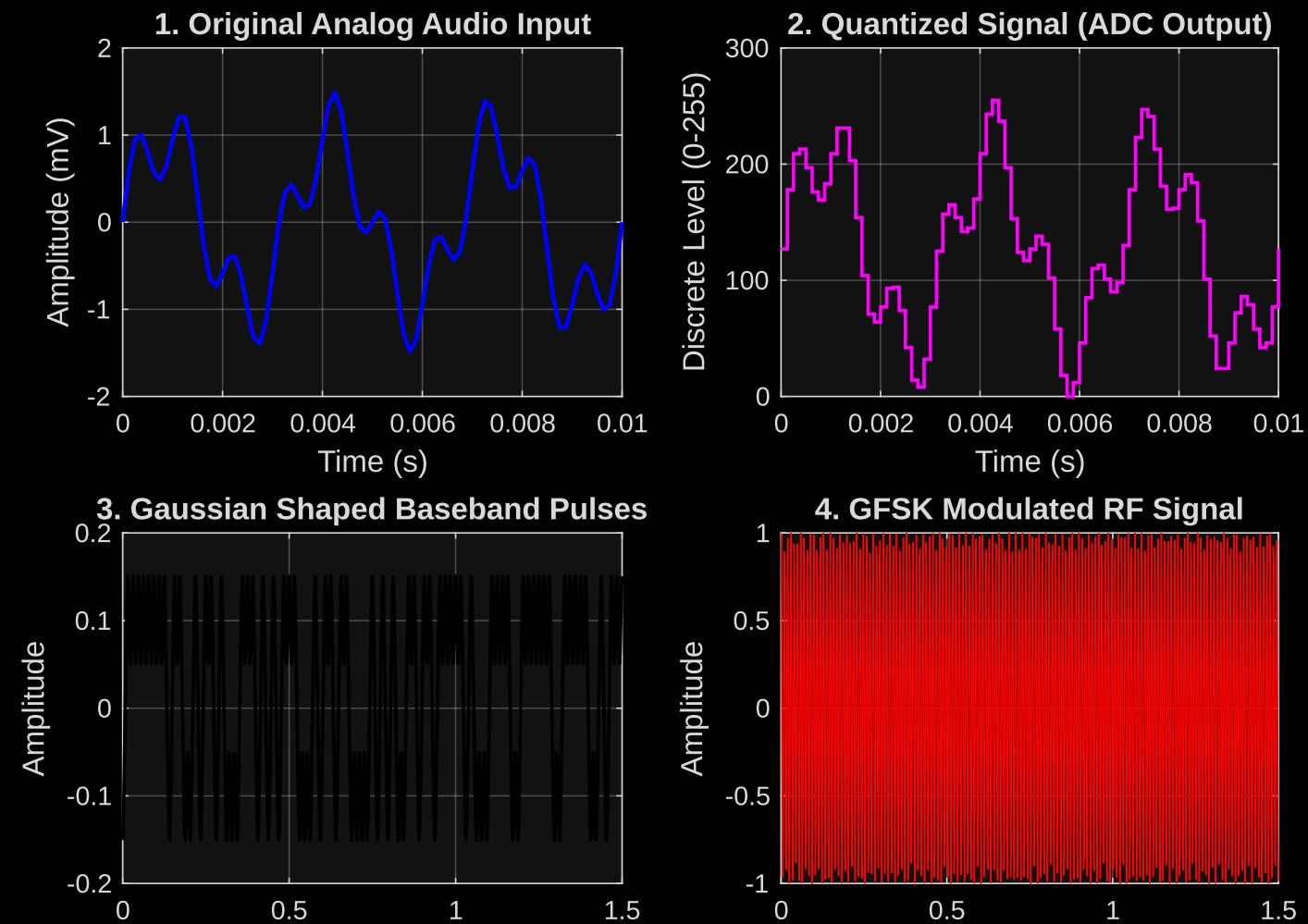
- **Gain Capacitor (10 μ F on Pin 1 & 8):** Bypasses the internal 1.35k Ω resistor to jump the gain from 20x to 200x.
- **Bypass Capacitor (0.1 μ F):** Filters the internal supply rail to prevent power supply noise from being amplified.
- **Output Coupling Capacitor (220 μ F):** Blocks DC voltage while allowing the AC audio signal to pass safely.
- **Zobel Network (10 Ω + 0.1 μ F):** Filters out high-frequency radio interference and prevents the amplifier from oscillating.

Why the LM386 was Chosen

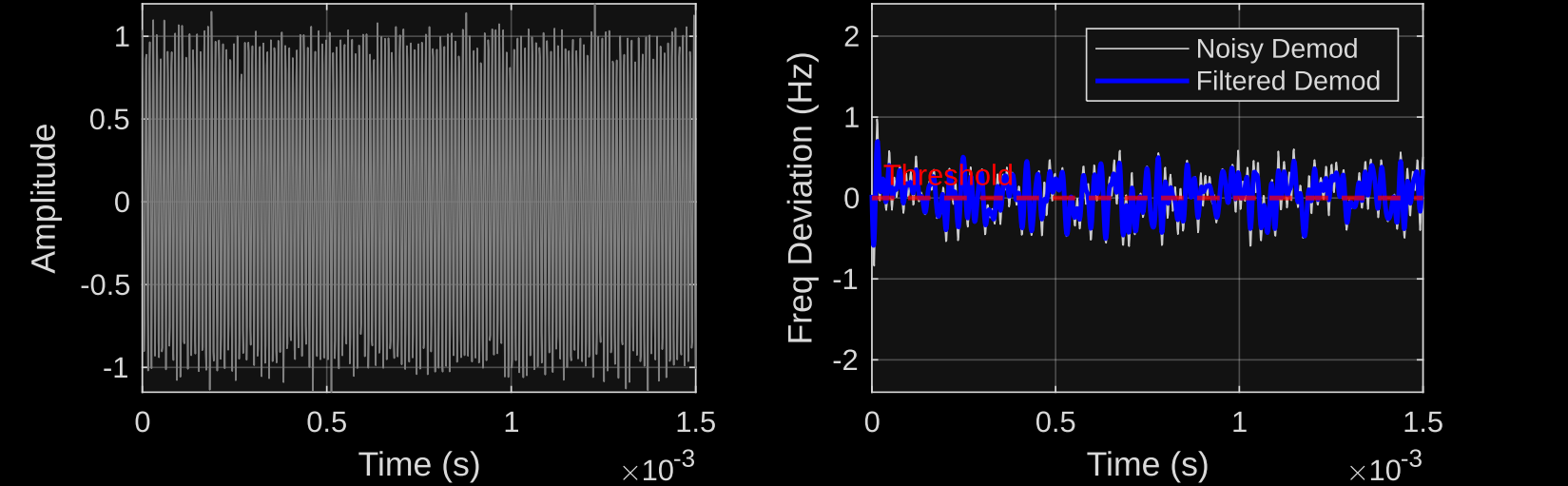
- **Wide Voltage Range:** Operates efficiently between 4V and 12V, making it ideal for battery-drained portable use.
- **Low Quiescent Current:** Draws very little power when no audio is playing, significantly extending battery life.
- **Self-Centering Output:** Automatically biases the output to half of VCC, reducing the need for complex external biasing resistors.

Mathematical Verification using MATLAB

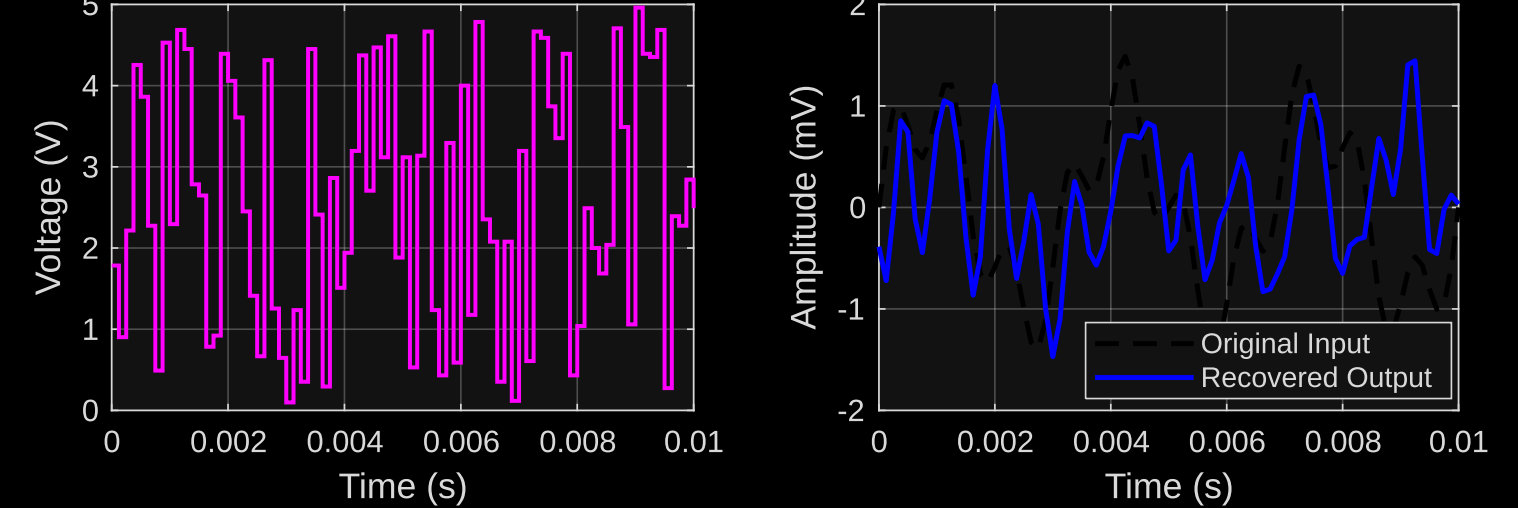
Digital Half-Duplex Voice Comm System Utilizing GFSK Modulation



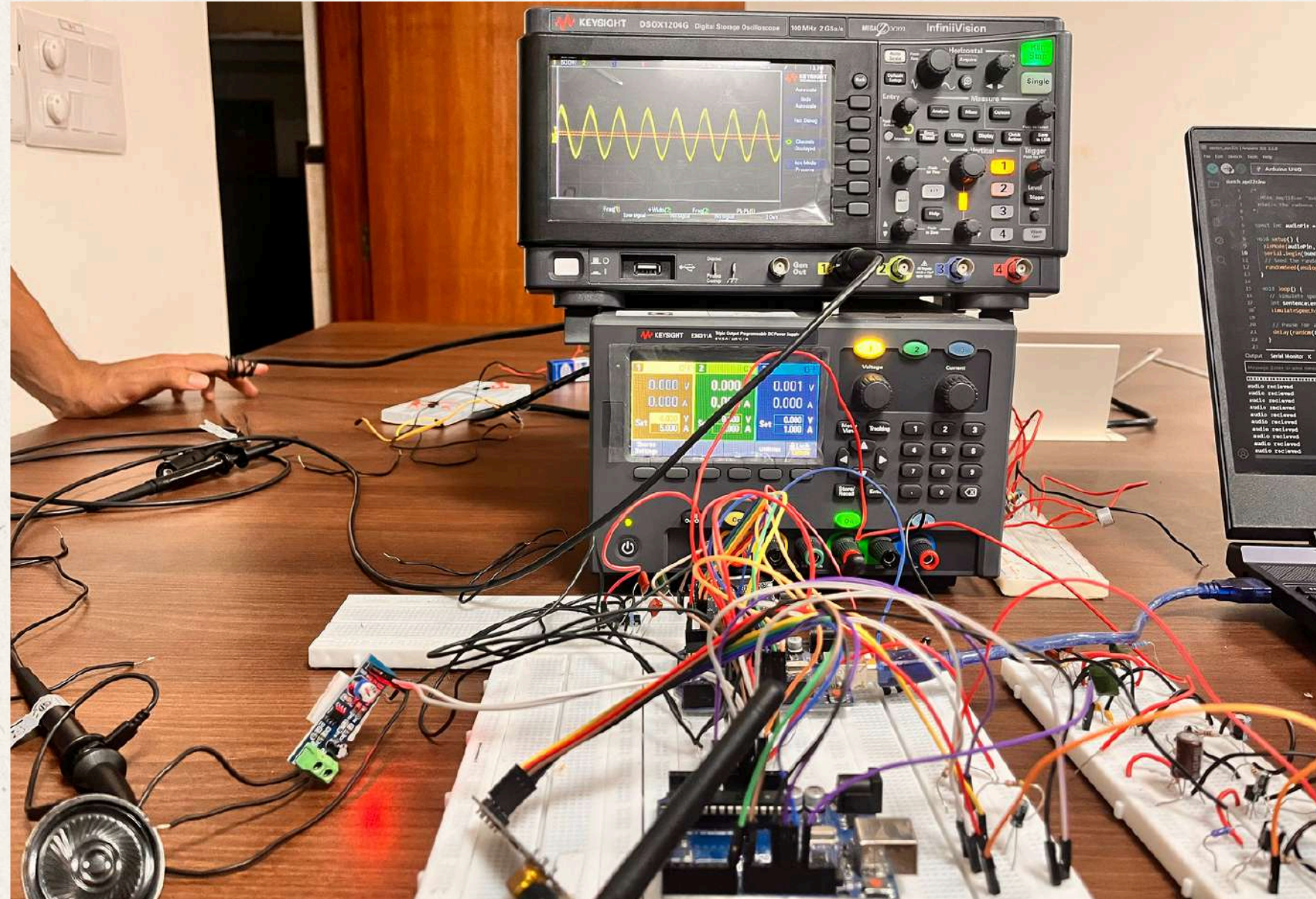
5. Received Signal with AWGN (SNR = 20dB) 6. Post-Detection Filtered Baseband (Inst. Freq)



7. DAC Output (Discrete Voltage Levels) 8. Final Recovered & Filtered Audio vs. Original



Result Verification



Comparison with Current Technology

Feature	GFSK Prototype (NRF24L01)	Consumer Analog (VHF/UHF FRS)	Commercial Digital (DMR)
Modulation	Digital (GFSK)	Analog (FM)	Digital (4FSK)
Operating Frequency	2.4 GHz (ISM Band)	462–467 MHz (UHF)	136–174 MHz or 400–470 MHz
Noise & EMI Immunity	Extremely High (Digital packets reject static; pristine audio near heavy machinery).	Low (Highly susceptible to electromagnetic interference, hum, and static).	Extremely High (Built with industrial-grade error correction).
Ideal Deployment Environment	Confined & Localized Spaces (Firms, institutional buildings, mines, dense factories).	Open Outdoor Areas (Line-of-sight environments, camping, hiking).	Wide-Area Critical Infrastructure (City-wide dispatch, massive industrial sites).
Signal Quality	High clarity within range; abruptly drops at the edge (no static).	Degrades gradually; progressively overwhelmed by static.	Perfect clarity; digital error correction maintains audio integrity.
Security / Privacy	High (Discrete digital payloads; easily encrypted via software).	Zero (Open analog broadcast; easily intercepted by any scanner).	Very High (Hardware-level AES-256 encryption).
Cost per Node	~₹1,400 (Customizable)	₹3,500 – ₹5,000	₹15,000+
Licensing	Globally License-Free	Generally License-Free	Requires FCC/WPC Licensing
Hardware Control	Absolute (Full open-source access to firmware and PCB logic).	None (Proprietary black-box hardware).	None (Locked proprietary ecosystems).

Estimated Cost

Component	Quantity	Estimated Unit Price (INR)	Estimated Total Price (INR)
Arduino UNO (Compatible Board)	2	₹200	₹400
NRF24L01 Transceiver Module	2	₹120	₹240
Electret Microphone	1	₹25	₹25
LM358 Operational Amplifier IC	1	₹15	₹15
LM386 Audio Amplifier Module	1	₹80	₹80
8 Ohm 0.5 W Speaker	1	₹40	₹40
Resistors, Capacitors, and Diodes (Assorted)	1 Set	₹50	₹50
General Purpose Transistors (e.g., BC547/2N2222)	1 Set	₹30	₹30
Power Supply (9V Battery with snap)	1	₹25	₹25
Total Estimated Cost			₹905

Sources

- Paper on "Performance analysis of GFSK modulation over AWGN channel" by Xiaoyue Zhang
- Paper on "Arduino based RF Compact Module for Short Range Wireless Communication using nRF24L0" by DR Anirban Karmakar, Tripura University
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THANK YOU!